Luolin Sun

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Research Profile

Physical oceanographer interested in geophysical fluid dynamics. I combine theoretical approaches and high-resolution/ensemble simulations to study ocean circulation, water-mass transformation, and tracer transport across scales. My recent work focuses on Subpolar to Arctic Ocean heat transport.

Research Experience

Postdoctoral Researcher, COAPS/Florida State University, USA 2024 – Present
Advisors: Prof. William Dewar (2024), Prof. Eric Chassignet (2025 – present)
Project 1: Impact of AMOC on ocean heat and freshwater transport to the Arctic.
Project 2: Ensemble-based diagnosis of surface eddy kinetic energy.
Responsibilities: Leading the two scientific projects and the preparation of associated manuscripts. Set up ocean model simulations (MITgcm and HYCOM) and numerical experiments.

Postdoctoral Researcher, IGE/CNRS, France 2023 – 2024
Advisors: Prof. William Dewar, Dr. Thierry Penduff
Project: Ensemble-based analysis of subtropical mode water formation
Responsibilities: Implemented a covariant coordinate system in MITgcm. Diagnosed ensemble-based potential vorticity and momentum budgets.

Education

PhD in Mathematics, Imperial College London, UK 2018 – 2022
Advisors: Prof. Pavel Berloff, Dr. Igor Shevchenko
Thesis: On the non-uniqueness of the mesoscale eddy diffusivity and its impact on mesoscale eddy parameterisation

MSci in Mathematics (First Class Honours), Imperial College London, UK 2014 – 2018
Thesis: Modelling stochastic uncertainty in the framework for Lagrangian particles

Publications

- **Sun, L.**, Uchida, T., Penduff, T., Dewar, W. K., Deremble, B., Poje, A. C., Chassignet, E. P., and Wienders, N. (2026). On the dynamics of the subtropical mode water from an ensemble view. **Journal of Physical Oceanography**, accepted. https://doi.org/10.1175/JPO-D-25-0115.1
- Poje, A. C., Uchida, T., Jamet, Q., **Sun, L.**, Penduff, T., Deremble, B., Schoonover, J., Trapanese, M., Wienders, N., and Dewar, W. K. (2026). Thoughts on prognostically modeling an eddying double-gyre ensemble mean. **Frontiers in Marine Science.** https://doi.org/10.3389/fmars.2026.1739607
- Uchida, T., Jamet, Q., Dewar, W., Deremble, B., Poje, A. C., and **Sun, L.** (2024). Imprint of Chaos on the Ocean Energy Cycle from an Eddying North Atlantic Ensemble. **Journal of Physical Oceanography**. https://doi.org/10.1175/JPO-D-23-0176.1

- **Sun, L.**, Haigh, M., Shevchenko, I., Berloff, P., and Kamenkovich, I. (2021). On non-uniqueness of the mesoscale eddy diffusivity. **Journal of Fluid Mechanics**. https://doi.org/10.1017/jfm.2021.472
- Haigh, M., **Sun, L.**, McWilliams, J., and Berloff, P. (2021). On eddy transport in the ocean. Part II: The advection tensor. **Ocean Modelling**. https://doi.org/10.1016/j.ocemod.2021.101845
- Haigh, M., **Sun, L.**, McWilliams, J., and Berloff, P. (2021). On eddy transport in the ocean. Part I: The diffusion tensor. **Ocean Modelling**. https://doi.org/10.1016/j.ocemod.2021.101831
- Kamenkovich, I., Berloff, P., **Sun, L.**, and Haigh, M. (2021). Complexity of mesoscale eddy diffusivity in the ocean. **Geophysical Research Letters**. https://doi.org/10.1029/2020GL091719
- Haigh, M., **Sun, L.**, Shevchenko, I., and Berloff, P. (2020). Tracer-based estimates of eddy-induced diffusivities. **Deep Sea Research Part I**. https://doi.org/10.1016/j.dsr.2020.103264

Manuscripts in Preparation

- **Sun, L.**, et al. Role of ocean heat transport on the increasing Arctic Ocean heat content in recent decades. In prep.
- **Sun, L.**, et al. Ensemble versus time averaging: perspectives on mesoscale eddy kinetic energy. In prep.

Selected Presentations

- Ensemble versus time averaging: perspectives on mesoscale geostrophic eddy kinetic energy in the North Atlantic: Ocean Sciences Meeting 2026
- Role of the AMOC in ocean heat transport from the subpolar North Atlantic to the Arctic during recent decades: Ocean Sciences Meeting 2026
- Eighteen Degree Water Dynamics Viewed from an Ensemble: EGU General Assembly 2024, EGU General Assembly 2025, DRAKKAR Workshop 2025

Awards

- **Doris Chen Merit Award**, 2020: awarded to top PhD candidates of the year
- **Roth's Scholarship**, 2019: awarded for outstanding academic performance for PhD candidates
- **Rector's Scholarship**, 2014: awarded for greatest academic potential for undergraduates

Technical Skills

- **Ocean Modelling**: Configuring and running simulations using MITgcm and HYCOM.
- **Programming skills**: Proficient in Python (xarray, NumPy, xgcm, Matplotlib), MATLAB and Fortran for numerical modeling and data analysis.
- **High-Performance Computing**: Skilled in MPI/OpenMP, UNIX/Linux and parallel computing.
- **Numerical Methods**: Designing PDE solvers for ocean models with irregular domains.